Finite Elements in Two Dimensions

Linear triangular elements, basis-function support, and a diffusion computation

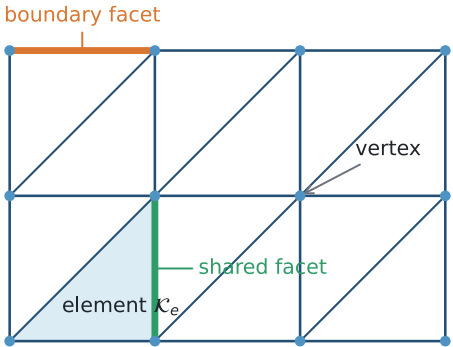
Lecture outline

Today we will connect four parts of the two-dimensional finite element method:

1. linear triangular elements and reference-to-physical mappings,
2. a diffusion problem with Dirichlet and Neumann boundary conditions,
3. global nodal basis functions and their support, and
4. a FEniCSx solution and an L^2 error study under mesh refinement.

We will state how local quantities enter $\boldsymbol{K}$ and $\boldsymbol{F}$, but we will not repeat the assembly derivation from Chapter 5. The complete multidimensional assembly procedure is in Chapter 6.

From intervals to triangular meshes



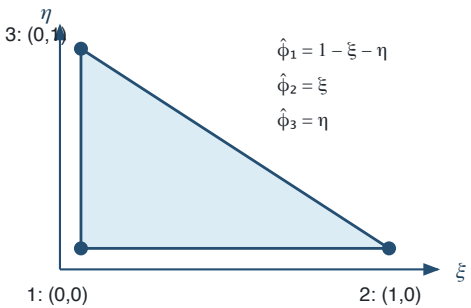
$I^e = (i_1, i_2, i_3)$

Connectivity lists the global vertices of each element.

Shared facets belong to two elements; boundary facets belong to one.

A triangular mesh partitions Ω into elements $\mathcal{K}_e$. Neighboring elements share vertices and edges, while boundary edges carry prescribed boundary data.

The reference P_1 triangle



$$\widehat{\mathcal{K}} = \{(\xi, \eta) : \xi \geq 0,\ \eta \geq 0,\ \xi + \eta \leq 1\}.$$

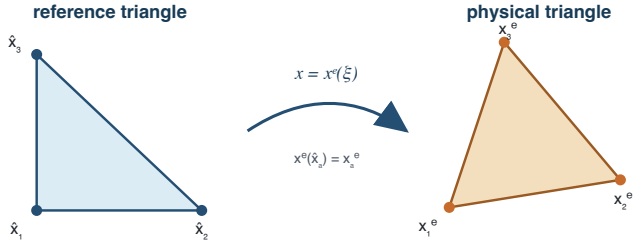
The nodal basis functions satisfy

$$\widehat{\phi}_a(\widehat{\boldsymbol{x}}_b) = \delta_{ab}, \qquad \sum_{a=1}^{3} \widehat{\phi}_a = 1.$$

Their gradients are constant on $\widehat{\mathcal{K}}$. Consequently, a P_1 triangle reproduces every affine scalar field exactly.

Isoparametric reference-to-physical mapping

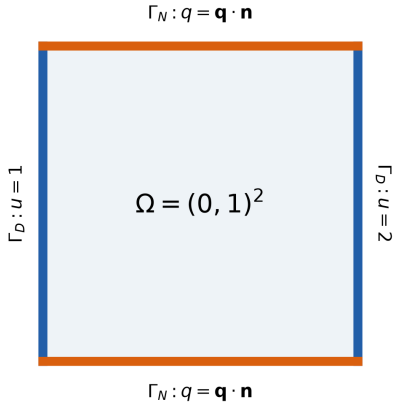
The geometry and finite element field use the same P_1 basis functions. This gives an **isoparametric mapping**.



$$\boldsymbol{x} = \boldsymbol{x}^e(\boldsymbol{\xi}) = \sum_{a=1}^{3} \widehat{\phi}_a(\boldsymbol{\xi}) \boldsymbol{x}_a^e, \qquad \boldsymbol{J}_e = \frac{\partial \boldsymbol{x}}{\partial \boldsymbol{\xi}}.$$

$$\nabla_{\boldsymbol{x}} \phi_a^e = \boldsymbol{J}_e^{-T} \nabla_{\boldsymbol{\xi}} \widehat{\phi}_a.$$

Diffusion problem on the unit square



Let $\Omega = (0,1)^2$. We solve

$$-\Delta u = f \quad \text{in } \Omega.$$

The vertical sides form

$$\Gamma_D = \{x = 0\} \cup \{x = 1\},$$
$$u = g \quad \text{on } \Gamma_D.$$

The horizontal sides form

$$\Gamma_N = \{y = 0\} \cup \{y = 1\},$$
$$\boldsymbol{q} = \nabla u, \qquad q = \boldsymbol{q} \cdot \boldsymbol{n} \quad \text{on } \Gamma_N.$$

For heat conduction, physical heat flow is $-\boldsymbol{q}$; positive q means heat enters the body.

Exact solution and problem data

Choose

$$u_\star(x, y) = 1 + x + x(1 - x)\sin(\pi y).$$

Substitution into the differential equation and boundary conditions gives

$$\begin{aligned} f(x, y) &= \left[2 + \pi^2 x(1 - x)\right]\sin(\pi y), \\ g(x, y) &= 1 + x && \text{on } \Gamma_D, \\ q(x, y) &= -\pi x(1 - x) && \text{on } \Gamma_N. \end{aligned}$$

Thus $u = 1$ on the left edge and $u = 2$ on the right edge. The prescribed normal flux is nonzero on the interiors of the top and bottom edges.

Variational and finite element problems

Define

$$\mathcal{U}_g = \{w \in H^1(\Omega) : w = g \text{ on } \Gamma_D\}, \qquad \mathcal{V}_0 = \{v \in H^1(\Omega) : v = 0 \text{ on } \Gamma_D\}.$$

The variational problem is

$$\text{find } u \in \mathcal{U}_g : \qquad \int_\Omega \nabla u \cdot \nabla v \, \mathrm{d}x = \int_\Omega fv \, \mathrm{d}x + \int_{\Gamma_N} qv \, \mathrm{d}s \quad \forall v \in \mathcal{V}_0.$$

The finite element problem restricts the trial and test functions to the continuous piecewise-linear space $\mathcal{V}_h$:

$$u_h \in \mathcal{U}_{h,g_h}, \qquad v_h \in \mathcal{V}_{h,0}.$$

The global algebraic system

Write the finite element field as

$$u_h = \sum_{j=1}^{N_n} U_j \phi_j.$$

Testing with the free nodal basis functions gives

$$\boldsymbol{K}\boldsymbol{U} = \boldsymbol{F},$$

with entries

$$K_{ij} = \int_\Omega \nabla\phi_j \cdot \nabla\phi_i \, \mathrm{d}x, \qquad F_i = \int_\Omega f\phi_i \, \mathrm{d}x + \int_{\Gamma_N} q\phi_i \, \mathrm{d}s.$$

In practice, the code computes element matrices $\boldsymbol{K}^e$, element load vectors $\boldsymbol{F}^e$, and boundary-edge contributions. Connectivity identifies the rows and columns of $\boldsymbol{K}$ and $\boldsymbol{F}$ that receive these local entries. Chapter 6 gives the complete assembly construction.

A global nodal basis function

Let $\boldsymbol{x}_i$ be mesh node i. The global basis function ϕ_i satisfies

$$\phi_i(\boldsymbol{x}_j) = \delta_{ij}.$$

It equals one at node i, equals zero at every other mesh node, and is linear on each triangle.
The incident-element set is

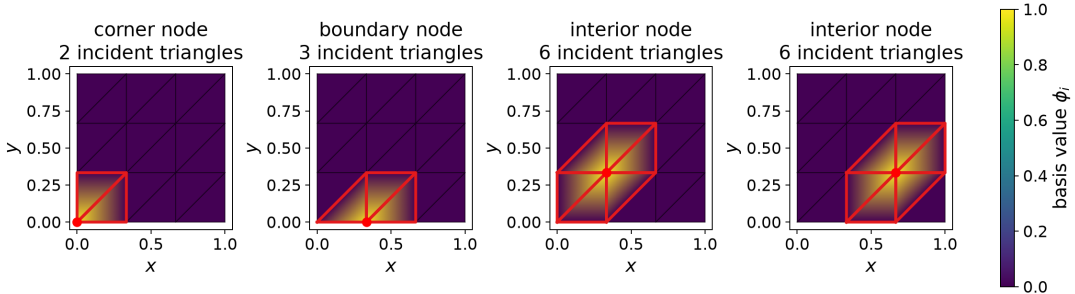
$$\mathcal{E}(i) = \{e : \boldsymbol{x}_i \text{ is a vertex of } \mathcal{K}_e\}.$$

Then

$$\operatorname{supp}(\phi_i) = \bigcup_{e \in \mathcal{E}(i)} \overline{\mathcal{K}_e}.$$

On every triangle outside this patch, all three nodal values of ϕ_i are zero, so the linear function is zero throughout that triangle.

Basis-function support on an 18-triangle mesh



Each plot sets one nodal coefficient equal to one and all others equal to zero. The highlighted patch consists exactly of the elements incident on the selected node.

Support and the stiffness matrix

The stiffness entry

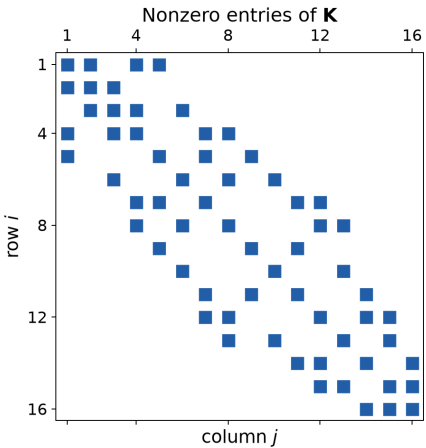
$$K_{ij} = \int_{\Omega} \nabla \phi_j \cdot \nabla \phi_i \, \mathrm{d}x$$

can be nonzero only where the supports of ϕ_i and ϕ_j overlap. Therefore,

$$\mathrm{supp}(\phi_i) \cap \mathrm{supp}(\phi_j) = \varnothing \quad \Longrightarrow \quad K_{ij} = 0.$$

For linear triangles, two nodal basis functions overlap only when their nodes belong to at least one common triangle. This local interaction produces a sparse global matrix even when the mesh contains many degrees of freedom.

Sparsity of the stiffness matrix



The 18-triangle mesh has 16 nodal degrees of freedom, so $\boldsymbol{K}$ is a 16×16 matrix.

A dot marks a nonzero entry K_{ij}. Nonzero off-diagonal entries occur only when nodes i and j belong to a common triangle.

Each node interacts with only a small part of the mesh. The resulting stiffness matrix is symmetric and sparse.

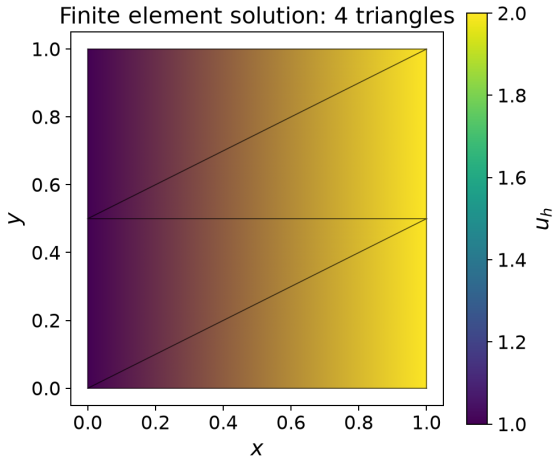
FEniCSx demonstration sequence

The live notebook follows the same mathematical order:

1. Create the triangular mesh and the continuous P_1 space.
2. Locate Γ_D and Γ_N and assign the boundary data.
3. Define the trial function, test function, and variational forms.
4. Solve the linear system for the nodal coefficients.
5. Compare u_h with $u_\star$ using the $L^2(\Omega)$ error.

We first use a separate 3×3 subdivision to inspect basis functions. The convergence study begins with four triangles and then halves the mesh size twice.

Solution on four triangles



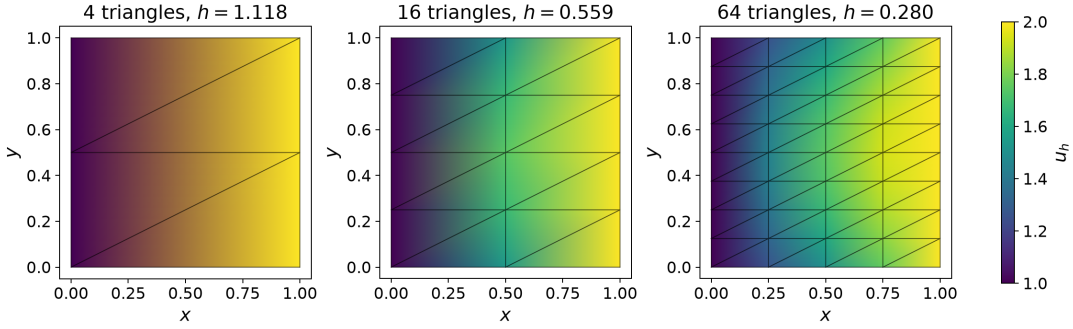
The mesh has six nodal degrees of freedom.

Every node lies on $x = 0$ or $x = 1$, where the prescribed values are 1 and 2. The coarse finite element field cannot yet represent the interior $x(1-x)\sin(\pi y)$ variation.

Measured error:

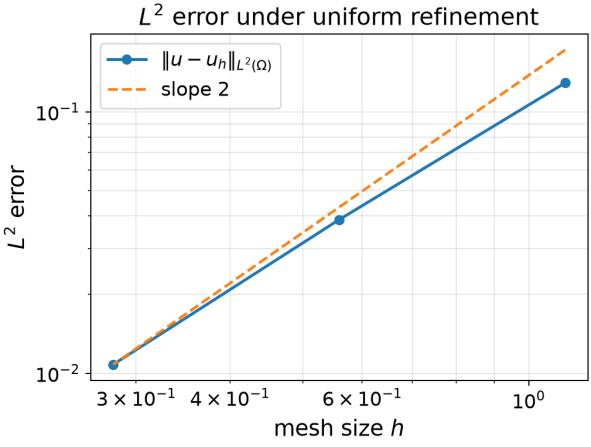
$$\|u_\star - u_h\|_{L^2(\Omega)} = 1.291 \times 10^{-1}.$$

Uniform mesh refinement



The three meshes contain 4, 16, and 64 triangles. Doubling the subdivisions in each coordinate changes the mesh size from h to $h/2$ and then to $h/4$.

L^2 error under refinement



triangles	h	L^2 error
4	1.118	1.291×10^{-1}
16	0.559	3.870×10^{-2}
64	0.280	1.079×10^{-2}

The observed rates are 1.74 and 1.84. The meshes are still coarse, but the error approaches the second-order L^2 behavior expected for smooth solutions and linear elements.

What to retain from today

- A two-dimensional P_1 function is linear on each triangle and continuous across shared edges.
- Reference basis functions and the element map provide basis values, physical gradients, and element integration.
- Each global nodal basis function is supported only on the triangles incident on its node.
- The discrete problem has the form $\boldsymbol{KU} = \boldsymbol{F}$; local matrices and vectors enter the global system through connectivity.
- Uniform refinement reduces the measured L^2 error for the diffusion problem.

Read Chapter 6 for the complete assembly procedure, quadrature rules, quadrilateral and three-dimensional elements, and vector-valued elasticity.